THE SQUARE OVER NESTED AFFINES OF A RELATIVE SPEC IS CARTESIAN

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Sections of a quasi-coherent algebra. schemes-lemma-widetilde-pullback (cross-chapter, named). R tensor_{R-prime} A-prime to A is an isomorphism.
- Fibre products of affines. schemes-lemma-fibre-product-affine-schemes (cross-chapter, named). Spec of a tensor product is the fibre product of the spectra.

Source reference. Original reference 01I4: focused-seeds/geometry/S9-cons-spec-inclusion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I9: focused-seeds/geometry/S9-cons-spec-inclusion/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LM: sources/stacks-constructions/pinned-original-source.tex, source lines 530–530. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01LM. Suppose $U \subset U' \subset S$ are affine opens. Let $A = \mathcal{A}(U)$ and $A' = \mathcal{A}(U')$. The map of rings $A' \to A$ induces a morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(A')$, and the diagram*

$$\begin{array}{ccc} \mathrm{Spec}(A) & \longrightarrow & \mathrm{Spec}(A') \\ \downarrow & & \downarrow \\ U & \longrightarrow & U' \end{array}$$

is cartesian.

Proof. Let $R = \mathcal{O}_S(U)$ and $R' = \mathcal{O}_S(U')$. Note that the map $R \otimes_{R'} A' \to A$ is an isomorphism as $\mathcal{A}$ is quasi-coherent (see Schemes, Lemma 01I9 for example). The result follows from the description of the fibre product of affine schemes in Schemes, Lemma 01I4. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.